DDL and Relational Algebra

The Foundation of SQL

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Data Definition Language (DDL)

Introducing SQL

- SQL, or Structured Query Language, is the standard language for communicating with relational databases.
- It is a **declarative language**, which is a major paradigm shift.

Declarative vs. Procedural
You specify **what** data you want, not **how** to get it. The DBMS figures out the most efficient way to execute your query. This is a huge advantage over file-based systems where you have to write the procedure yourself.

The Sub-Languages of SQL

SQL is composed of several sub-languages, each with a specific purpose.

Core Languages

- **DDL** (Data Definition Language)
- **DML** (Data Manipulation Language)
- **DQL** (Data Query Language)

Other Languages

- **DCL** (Data Control Language)
- **TCL** (Transaction Control Language)

Data Definition Language (DDL)

- DDL is a subset of SQL used to define the database structure or schema.
- It deals with creating, altering, and dropping database objects.
- Common DDL commands:
  - `CREATE`
  - `ALTER`
  - `DROP`
  - `TRUNCATE`

The `CREATE TABLE` Statement

The `CREATE TABLE` statement is used to create a new table in a database.

Syntax

```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype,
    ...
);
```

The CREATE TABLE Statement

Example

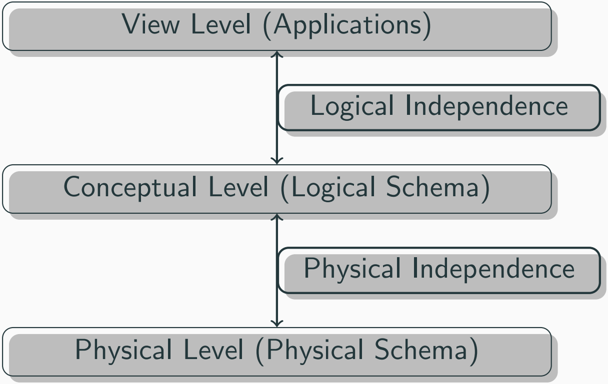
```
CREATE TABLE Students (
    StudentID INT,
    FirstName VARCHAR(255),
    LastName VARCHAR(255)
);
```

Common SQL Data Types

- **Numeric Types:**
  - `INT` or `INTEGER`: For whole numbers.
  - `DECIMAL(p,s)`: For fixed-point numbers. ‘p‘ is total digits, ‘s‘ is digits after decimal.
  - `FLOAT`, `REAL`: For floating-point numbers.
- **String Types:**
  - `CHAR(n)`: Fixed-length string of ‘n‘ characters.
  - `VARCHAR(n)`: Variable-length string up to ‘n‘ characters.
  - `TEXT`: For long-form text.
- **Date and Time Types:**
  - `DATE`: Stores year, month, and day.
  - `TIME`: Stores hour, minute, and second.
  - `TIMESTAMP`: Stores date and time.

A Key Concept: Physical Data Independence

- **Definition:** The ability to modify the physical schema without causing application programs to be rewritten.
- The conceptual schema insulates applications from changes in physical storage.



CREATE TABLE with Constraints

Constraints are rules enforced on the data columns of a table.

- PRIMARY KEY: Uniquely identifies each record.
- FOREIGN KEY: Links two tables together.
- NOT NULL: Ensures a column cannot have a NULL value.
- UNIQUE: Ensures that all values in a a column are different.
- CHECK: Ensures that all values in a column satisfy a specific condition.

Example

```
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    FirstName VARCHAR(255) NOT NULL,
    Email VARCHAR(255) UNIQUE
);
```

Relational Algebra

Introduction to Relational Algebra

- Relational algebra is a **procedural query language**.
- It takes instances of relations as input and yields instances of relations as output.
- It consists of a set of operators that are used to form queries.
- It is the theoretical foundation of SQL.

Types of Operators

- **Unary Operators:** Operate on a single relation.
    - Selection (σ)
    - Projection (Π)
- **Binary Operators:** Operate on pairs of relations.
    - Union ($\cup$)
    - Intersection ($\cap$)
    - Set Difference ($-$)
    - Cartesian Product ($\times$)

Selection (σ)

- Selects tuples (rows) that satisfy a given predicate.
- Syntax: $\sigma_{predicate}(R)$
- SQL equivalent: `WHERE` clause.

Selection Example

Query: Select all students with a ‘Discipline‘ of ’CSE’.

Students

ID	Name	Discip
101	Parth	CSE
102	Juhi	DSAI
103	Isha	CSE

$\xrightarrow{\sigma_{Discip='CS'}}$

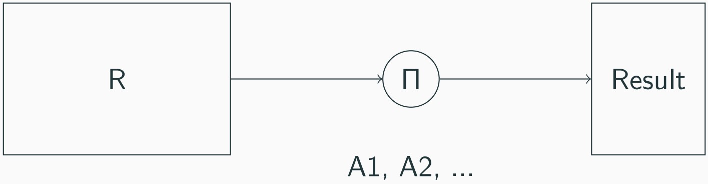
Result

ID	Name	Discip
101	Parth	CSE
103	Isha	CSE

SQL: `SELECT * FROM Students WHERE Discip = ’CS’;`

Projection (Π)

- Selects certain columns (attributes) from a relation.
- Syntax: $\Pi_{A1,A2,\ldots,An}(R)$
- SQL equivalent: `SELECT` clause.



Projection Example

Query: Project the 'Discipline' of all students. Note the duplicate removal.

Students

ID	Name	Discip
101	Parth	CSE
102	Juhi	DSAI
103	Isha	CSE

$\xrightarrow{\Pi_{Discip}}$

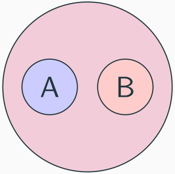
Result

Discip
CSE
DSAI

SQL: `SELECT DISTINCT Discip FROM Students;`

Union (∪)

- Returns a relation containing all tuples that appear in either or both of two relations.
- The two relations must be **union-compatible** (same number of attributes, and corresponding attributes have the same domain).
- SQL equivalent: `UNION`.



Union Example

Query: List all students who are in either CSE or DSAI.

CSE

ID	Name
101	Parth
103	Isha

DSAI

ID	Name
102	Juhi
104	Suman

$\cup$

Result

ID	Name
101	Parth
103	Isha
102	Juhi
104	Suman

Union Example

Query: List all students who are in either CSE or DSAI.

SQL: `SELECT * FROM CSE UNION SELECT * FROM DSAI;`

CSE

ID	Name
101	Parth
103	Isha

DSAI

ID	Name
102	Juhi
104	Suman

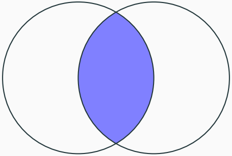
∪

Result

ID	Name
101	Parth
103	Isha
102	Juhi
104	Suman

Intersection ($\cap$)

- Returns a relation containing all tuples that appear in **both** of two relations.
- The two relations must be union-compatible.
- SQL equivalent: `INTERSECT`.



Query: Find students who are members of both the Coding Club and the Robotics Club.

Coding_Club

ID	Name
101	Parth
102	Juhi
103	Isha

Robotics_Club

ID	Name
102	Juhi
104	Suman

$\cap$

Result

ID	Name
102	Juhi

Intersection Example

Query: Find students who are members of both the Coding Club and the Robotics Club.

SQL: `SELECT * FROM Coding_Club INTERSECT SELECT * FROM Robotics_Club;`

Coding_Club

ID	Name
101	Parth
102	Juhi
103	Isha

Robotics_Club

ID	Name
102	Juhi
104	Suman

$\cap$

Result

ID	Name
102	Juhi

Thank You!!

Questions?